

Model Evaluation Summary

Financial Loan Default Prediction Project

Approach

Two classification models were trained and compared: Logistic Regression and Random Forest, both using `class_weight='balanced'` to address class imbalance, since only 20% of loans in the dataset were actual defaults.

Results

Metric	Logistic Regression	Random Forest
Accuracy	67%	80%
Recall (Default)	65%	14%
Precision (Default)	34%	51%
F1-Score (Default)	0.45	0.21

Key Finding

Although Random Forest achieved a higher overall accuracy (80%), its recall for the default class dropped sharply to 14%, indicating it failed to identify most actual defaulters. Logistic Regression, despite a lower accuracy of 67%, achieved a substantially higher recall of 65%, making it significantly more effective at identifying risky borrowers.

Conclusion

Accuracy alone proved to be a misleading metric due to the class imbalance in the dataset, since high accuracy could be achieved simply by predicting the majority class. Logistic Regression was selected as the final model, as it aligns better with the core business objective of correctly identifying loan default risk, prioritizing recall over raw accuracy.